

# **Scientific Project**

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# 1 Introduction

## 1.1 Research Motivation

Spectral analysis has proven to be a powerful tool for solving a wide range of problems in mathematics, physics and engineering. Among its many applications, it facilitates the decoupling of problems involving vectors or functions into collections of simpler problems involving only scalars, the analysis of the phenomenon of resonance, and the characterization of the asymptotic behavior of a system [7]. Therefore, the ability to compute the spectra of linear operators is of fundamental importance.

In practice, however, analytical solutions are rarely available. Even in the case of finite-dimensional operators represented by  $n \times n$  matrices, the theorem of Abel and Ruffini shows that no general formula, involving only rational operations and root extractions, can exist for solving algebraic equations of degree higher than fourth [8]. As a result, there is no general closed-form expression for the eigenvalues of arbitrary matrices, making numerical methods necessary.

The situation becomes even more challenging when considering infinite-volume operators or those acting on infinite-dimensional spaces. Nevertheless, their ubiquity in applications has driven the development of multiple numerical algorithms designed to approximate the spectra for different classes of operators.

Motivated by these challenges, the authors of [4] address the problem of computing the spectra for *short-range finite-local-complexity infinite-volume operators*. Prior to their work, no numerical method was capable of approximating the spectra for this specific class of operators with full error control. By overcoming this limitation, their work fills an important gap in the existing literature.

## 1.2 Problem Statement

The broad aim of this work is to validate and contribute to the understanding of some of the results reported in [4] in concrete situations. To this end, simulation tools which take advantage of extensively-tested scientific libraries such as NumPy and SciPy were developed in Python. This allowed us to fully focus on the mathematical aspects of the problem, and on the interpretation of the results, without having to worry about the technical details of the implementation.

## 2 Theoretical Framework

### 2.1 Basic Definitions

#### 2.1.1 Metric and open balls on $\mathbb{R}^n$

In this dissertation, unless otherwise stated, we define the distance  $d$  on  $\mathbb{R}^n$  as the *maximum distance*

$$d(x, y) = d((x_1, \dots, x_n), (y_1, \dots, y_n)) := \max_{k=1, \dots, n} |x_k - y_k| \quad (1)$$

for all  $x, y \in \mathbb{R}^n$ .

We also define the set  $B_r(x)$  for  $r > 0$  and  $x \in \mathbb{R}^n$  as the open ball using this distance, i.e. as the hypercube of side length  $2r$  centered at  $x$ .

#### 2.1.2 Uniformly Discrete Set

**Definition.** A subset  $\Gamma \subseteq \mathbb{R}^n$  is called *uniformly discrete* if there exists a constant  $q > 0$ , the *packing radius*, such that  $d(x, y) \geq q$  for all  $x, y \in \Gamma$  with  $x \neq y$ .

#### 2.1.3 Discrete Operator and Matrix Elements

**Definition.** A *discrete operator*  $H$  in dimension  $n \in \mathbb{N}$  is a bounded operator on a separable Hilbert space  $\mathcal{H}$ , together with an orthonormal basis  $(e_i)_{i \in \Gamma}$  indexed by a uniformly discrete set  $\Gamma \subset \mathbb{R}^n$ .

**Remark.** The orthonormal basis  $(e_i)_{i \in \Gamma}$  should be thought of as a collection of *delta functions* supported on the points of  $\Gamma$ : each basis vector  $e_x$  is localized at a single point  $x \in \Gamma$ , and satisfies  $\langle e_x, e_y \rangle = \delta_{xy}$ .

We also define the *matrix elements* of a discrete operator  $H$  at the points  $x, y \in \Gamma$  as  $H_{xy} = \langle e_x, H e_y \rangle$ .

We shall always represent discrete operators with respect to the special basis  $(e_i)_{i \in \Gamma}$  and identify  $\mathcal{H}$  with  $\ell^2(\Gamma)$  through the basis isomorphism.

#### 2.1.4 Finite-Dimensional Subspaces and Orthogonal Projections

For any  $x \in \mathbb{R}^n$  and  $r > 0$ , the finite-dimensional subspace  $\mathcal{H}_{B_r(x)} \subset \mathcal{H}$  is defined by  $\mathcal{H}_{B_r(x)} := \text{span}\{e_y | y \in B_r(x) \cap \Gamma\}$ , and the orthogonal projection onto  $\mathcal{H}_{B_r(x)}$  is denoted by  $\mathbf{1}_{B_r(x)}$ .

#### 2.1.5 Short- and Finite-Range Operators

**Definition.** Let  $H$  be a discrete operator in dimension  $n$ . Then  $H$  is called *short-range* if there exists  $C$  and  $\varepsilon > 0$  such that

$$|H_{xy}| \leq C d(x, y)^{-(n+\varepsilon)} \quad (2)$$

for all  $x, y \in \Gamma$ .

**Definition.** A discrete operator  $H$  is said to have *finite range* if there exists a constant  $m > 0$  (the maximal hopping length) such that  $H_{xy} = 0$  whenever  $d(x, y) > m$ .

### 2.1.6 Equivalent Action and Finite Local Complexity

**Definition.** A discrete operator  $H$  is said to have an *equivalent action* on two subsets  $A, B \subseteq \Gamma$  if there is  $t \in \mathbb{R}^n$  such that  $B = A + t$  and if there exists a  $U : A \rightarrow S^1 \subset \mathbb{C}$ ,  $z \mapsto U(z)$  such that for any  $a_1, a_2 \in A$  we have

$$H_{b_1 b_2} = U(a_1) H_{a_1 a_2} U(a_2)^* \quad (3)$$

where  $b_1 = a_1 + t$ ,  $b_2 = a_2 + t$ .

**Remark.** Note that for any operator  $H$ , the equivalent action of  $H$  defines an equivalent relation on subsets of  $\Gamma$ .

**Definition.** A discrete operator  $H$  is said to have *finite local complexity* (FLC) if for any  $r > 0$ , the set  $\{\Gamma \cap B_r(x) | x \in \mathbb{R}^n\}$  can be partitioned into finitely many equivalence classes with respect to the equivalent action of  $H$ .

### 2.1.7 Hausdorff Distance

**Definition.** Let  $(X, d)$  be a metric space, and  $\mathcal{F}(X)$  the set of all nonempty bounded closed sets in  $X$ . For  $(A, B) \in \mathcal{F}(X) \times \mathcal{F}(X)$ , the *Hausdorff distance* between  $A$  and  $B$  is defined as [2]

$$d_{\text{Hausdorff}}(A, B) := \max \left\{ \sup_{a \in A} d(a, B), \sup_{b \in B} d(A, b) \right\} \quad (4)$$

where  $d(x_0, E) := \inf_{e \in E} d(x_0, e)$  is the distance of a point  $x_0 \in X$  to a nonempty subset  $E \subseteq X$ .

## 2.2 The Spectral and Pseudospectral Problems

Next, we give some additional definitions relevant to the discussion of error bounds on the approximant spectra.

### 2.2.1 Normal Operators

**Definition.** Let  $A$  be a linear operator on an inner product space  $V$ . Then  $A$  is *normal* if and only if  $AA^* = A^*A$ .

**Remark.** All self-adjoint and all isometric linear operators are normal [3].

### 2.2.2 The Pseudospectrum

From mechanical oscillations to oscillations in quantum mechanics, most of the familiar applications of spectrum analysis involve matrices or operators that are normal or close to normal. Yet this familiarity can be deceiving: for nonnormal matrices or operators, the spectrum alone may tell us surprisingly little. By way of example, if a system is modeled by a linear normal operator, then the resonant amplification experienced by it in response to a sinusoidally varying driving force with frequency  $\omega$  will be inversely proportional to the distance in the complex plane between  $\omega$  and the nearest eigenvalue. However, for a nonnormal system, the resonant amplification may be orders of magnitude greater. The resonances are simply not determined by the eigenvalues alone [7]. It is precisely this inadequacy that motivates the pseudospectrum.

**Definition.** Let  $H$  be a linear operator with spectrum  $\sigma(H)$ , then the map  $\rho_H : \mathbb{C} \rightarrow \mathbb{R}_+$  defined as

$$\rho_H(z) := \begin{cases} \|(H - z)^{-1}\|^{-1} & \text{for } z \notin \sigma(H) \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

is called the *lower norm function*.

Although  $\rho_H$  is not a norm, it can be given a variational definition, similar to the operator norm.

**Definition (Variational).** Let  $H$  be a linear operator on a Hilbert space  $\mathcal{H}$ , then

$$\rho_H(z) := \inf_{\substack{\psi \in \mathcal{H} \\ \|\psi\|=1}} \|(H - z)\psi\| \quad (6)$$

is the *lower norm function*.

Furthermore,  $\rho_H$  is Lipschitz continuous with Lipschitz constant equal to 1.

**Definition.** Let  $H$  be a linear operator, then the  $\varepsilon$ -pseudospectrum is defined as

$$\sigma_\varepsilon(H) := \{z \in \mathbb{C} \mid \rho_H(z) \leq \varepsilon\} \quad (7)$$

**Proposition.** Let  $H$  be a linear operator with spectrum  $\sigma(H)$ , then

$$\rho_H(z) \leq d(z, \sigma(H)) \quad (8)$$

with normality of  $H$  being sufficient for equality.

Thus, the  $\varepsilon$ -pseudospectrum of a normal operator is simply the  $\varepsilon$ -fattening of its spectrum.

**Remark.** Pseudospectral analysis offers a more informative object for studying the behavior of operators for which the spectrum alone is insufficient. In particular, it shifts the focus from exact eigenvalues to  $\varepsilon$ -approximate ones, which, as H. J. Landau observed [5], carry over smoothly from the normal case:

The distinction between them arises, and plays a crucial role, in passing to non-Hermitian equations where, as we have seen, true eigenvalues do not in general describe the operator adequately; on the other hand ... systems of mutually orthogonal  $\varepsilon$ -approximate eigenfunctions are easily constructed, and the behavior of  $\varepsilon$ -approximate eigenvalues, as well as their role in explaining the action of the operator, carry over smoothly from the Hermitian case.

**Theorem.** Let  $H$  be a closed operator on a complex Banach space, then the pseudospectra  $\{\sigma_\varepsilon(H)\}_{\varepsilon>0}$  have the following properties [7].

- Each  $\sigma_\varepsilon(H)$  is a nonempty open subset of  $\mathbb{C}$ , and any bounded connected component of  $\sigma_\varepsilon(H)$  intersects the spectrum  $\sigma(H)$ .
- The pseudospectra are strictly nested supersets of the spectrum:  $\bigcap_{\varepsilon>0} \sigma_\varepsilon(H) = \sigma(H)$ , and conversely, for any  $\delta > 0$ ,  $\sigma_{\varepsilon+\delta}(H) \supseteq \sigma_\varepsilon(H) + \Delta_\delta$ , where  $\Delta_\delta$  is the open disk of radius  $\delta$  and  $+$  is the Minkowski sum.

## 2.3 Approximating the (Pseudo-)Spectrum with Full Error Control

As shown in [4], as long as we are dealing with operators with finite local complexity, it is always possible to compute the spectrum in the normal case, and the pseudospectrum in the nonnormal one. In this section we provide the theorems that led to these results.

### 2.3.1 The Pseudospectral Inclusion Bound (PIB)

An upper bound on  $\rho_H$  can be obtained by performing a singular value decomposition on the rectangular matrix  $(H_{yz})_{y \in B_{r+m}(x), z \in B_r(x)}$ , where  $x \in \mathbb{R}^n$  and  $r, m > 0$  are fixed numbers.

**Remark.** Recall that a *singular value decomposition* (SVD) of a matrix  $A \in \mathbb{C}^{m \times n}$  is a factorization into a product of matrices  $A = USV^*$ , where  $U \in \mathbb{C}^{m \times m}$  and  $V \in \mathbb{C}^{n \times n}$  are unitary, and  $S \in \mathbb{C}^{m \times n}$  is diagonal. The matrix  $S$  contains the square roots of the nonnegative eigenvalues of both  $AA^*$  and  $A^*A$ , and these are the *singular values* of  $A$  [6].

**Definition.** Let  $H$  be a discrete operator with finite range  $m > 0$ , let  $x \in \mathbb{R}^n$  be any point,  $r > 0$  and  $\lambda \in \mathbb{C}$ . The *uneven section* for these data is the linear operator

$$\begin{aligned} Q_{r,\lambda,x} : \mathcal{H}_{B_r(x)} &\rightarrow \mathcal{H}_{B_{r+m}(x)} \\ Q_{r,\lambda,x}(\psi) &= \mathbf{1}_{B_{r+m}(x)}(H - \lambda)\psi \end{aligned}$$

Since  $\mathcal{H}_{B_r(x)}$  and  $\mathcal{H}_{B_{r+m}(x)}$  are finite dimensional, we can represent  $Q_{r,\lambda,x}$  as a rectangular matrix.

**Theorem.** Let  $H$  be a discrete operator with finite range  $m > 0$ , let  $x \in \mathbb{R}^n$  be any point,  $r > 0$  and  $\lambda \in \mathbb{C}$ . Let  $\varepsilon_{r,\lambda,x}$  be the smallest singular value of  $Q_{r,\lambda,x}$ . Then  $\lambda$  is contained in the  $\varepsilon_{r,\lambda,x}$ -pseudospectrum of  $H$ , that is,

$$\rho_H(\lambda) \leq \varepsilon_{r,\lambda,x} \tag{9}$$

A proof can be found in [4].

### 2.3.2 The Spectral Gap Bound

In previous work [1], it was shown that the pseudospectral inclusion bound  $\varepsilon_{r,\lambda,x}$  always converges to  $\rho_H(\lambda)$  as  $r \rightarrow \infty$ , at least for normal, finite-range operators. Nonetheless, since the convergence is not uniform in the operator  $H$ , it cannot be used to compute a lower bound on  $\rho_H(\lambda)$ .

One way to circumvent this is by making use of the infimum of  $\varepsilon_{r,\lambda,x}$  over all  $x \in \mathbb{R}^n$ . More precisely, for

$$\varepsilon_{r,\lambda} := \inf_{x \in \mathbb{R}^n} \varepsilon_{r,\lambda,x}$$

we get the following theorem.

**Theorem.** Let  $n \in \mathbb{N}$ , let  $\Gamma \subseteq \mathbb{R}^n$  be a uniformly discrete set with packing radius  $q > 0$ , let  $H$  be a discrete operator on  $\ell^2(\Gamma)$  with finite range  $m > 0$  and let  $M := \sup_{x,y \in \Gamma} |H_{xy}|$ . Then for every  $\lambda \in \mathbb{C}$  and every  $r > m$  it holds that

$$\rho_H(\lambda) \geq \varepsilon_{r,\lambda} - \frac{C}{r} \tag{10}$$

where

$$C := mM \left( \frac{36m}{q} \right)^{n/2} \quad (11)$$

For a proof of this theorem see [4].

In practice, the importance of this theorem lies in the fact that for FLC operators, the infimum  $\varepsilon_{r,\lambda}$  can be computed by evaluating  $\varepsilon_{r,\lambda,x}$  for a finite number of suitably chosen points  $x \in \mathbb{R}^n$ .

### 2.3.3 A General Algorithm

Finally, combining bounds 9 and 10 leads to a general algorithm for computing an approximation to the (pseudo-)spectrum for FLC operators with full error control. That is, given any short-range, discrete, normal, FLC operator  $H$  and  $k \in \mathbb{N}$ , one can derive an algorithm which computes an approximation  $\Gamma_k(H) \subseteq \mathbb{C}$  such that

$$d_{\text{Hausdorff}}(\sigma(H), \Gamma_k(H)) \leq 2^{-k} \quad (12)$$

Similarly, for nonnormal operators one can derive an algorithm which computes for any  $\varepsilon > 0$  and approximation  $\Gamma_k(H, \varepsilon)$  such that

$$d_{\text{Hausdorff}}(\sigma_\varepsilon(H), \Gamma_k(H, \varepsilon)) \leq 2^{-k} \quad (13)$$

For a proof of either of these results, see [4].



### 3 Results in Concrete Cases

#### 3.1 The Free Hamiltonian

As our first case, we study the discretization of the time-independent one-dimensional free Schrödinger equation with periodic boundary conditions.

The finite difference method for the boundary-value problem,

$$H_{\text{free}}\psi(x) = -\frac{1}{2}\frac{d^2}{dx^2}\psi(x) = E\psi(x), \quad \text{for } 0 \leq x \leq L, \quad \text{with } \psi(0) = \psi(L)$$

requires that we select an integer  $N > 0$  and divide the interval  $[0, L]$  into  $(N - 1)$  equal subintervals whose endpoints are the mesh points  $x_i = i\Delta x$ , for  $i = 0, 1, 2, \dots, N - 1$ , where  $\Delta x = L/(N - 1)$ . Using a centered-difference formula for  $\psi''(x_i)$  together with the boundary conditions, we get the system of equations

$$-\frac{1}{2}\frac{1}{(\Delta x)^2}(\psi_{i+1} - 2\psi_i + \psi_{i-1}) = E\psi_i, \quad \text{for } i = 1, 2, \dots, N - 2;$$

$$\text{and } \psi_0 = \psi_{N-1}.$$

This allows us to express the Hamiltonian,  $H_{\text{free}} \in \mathcal{L}(\ell^2(\mathbb{Z}))$ , in matrix form as

$$H_{\text{free}} = \begin{pmatrix} 1 & -\frac{1}{2} & & & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} & & \\ & -\frac{1}{2} & 1 & & \\ & & & \ddots & \\ & & & & 1 & -\frac{1}{2} \\ -\frac{1}{2} & & & & -\frac{1}{2} & 1 \end{pmatrix}$$

for  $\Delta x = 1$ .

We can then plot its spectrum using a kernel density estimation (KDE). The resulting plot is shown in figure 1.

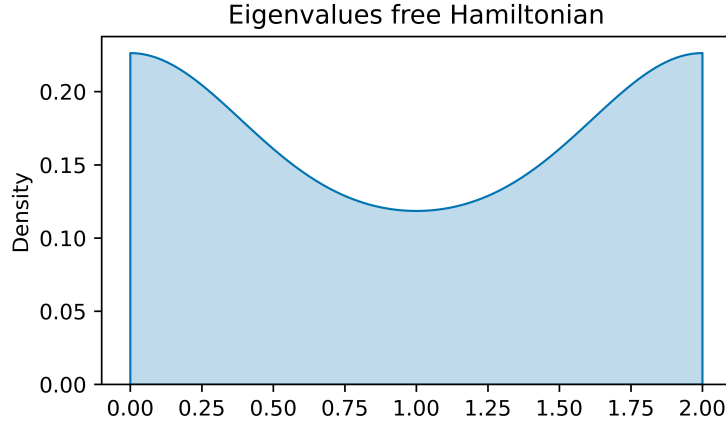


Figure 1: KDE plot of the spectrum of the free Hamiltonian, with  $L = 1000$  and  $\Delta x = 1$ .

Now, in order to validate the pseudospectral inclusion bound for this operator, first we need to take an uneven section. In fact, given the simplicity of  $H_{\text{free}}$ , we can center it at any point. For instance, let  $r \in \{50, 100, 150\}$ ,  $\lambda \in \mathbb{R}$ , and  $x = 500$ , then an uneven section looks like so

$$(H_{\text{free}} - \lambda) = \begin{pmatrix} 1 - \lambda & -\frac{1}{2} & & \dots & & -\frac{1}{2} \\ -\frac{1}{2} & 1 - \lambda & & & & \\ & & \ddots & & & \\ & & & 1 - \lambda & & \\ & & & -\frac{1}{2} & & \\ \vdots & & & & \boxed{\begin{matrix} -\frac{1}{2} & & & Q_{r,\lambda,x} \\ 1 - \lambda & -\frac{1}{2} & & \\ -\frac{1}{2} & 1 - \lambda & -\frac{1}{2} & \\ & \ddots & \ddots & \ddots \\ & & -\frac{1}{2} & 1 - \lambda \\ & & & -\frac{1}{2} \end{matrix}} & & \\ & & & & 1 - \lambda & \\ & & & & & \ddots \\ -\frac{1}{2} & & & \dots & & -\frac{1}{2} & 1 - \lambda \end{pmatrix}$$

And, for different  $\lambda$  values, we get the following results:

| Pseudospectral Inclusion Bound (PIB) |                         |                               |
|--------------------------------------|-------------------------|-------------------------------|
| $(r, \lambda)$                       | $d(\lambda, \sigma(H))$ | $\varepsilon_{r,\lambda,500}$ |
| (50, -0.1)                           | 0.100000                | 0.100507                      |
| (100, -0.1)                          | 0.100000                | 0.100125                      |
| (150, -0.1)                          | 0.100000                | 0.100055                      |
| (50, 0.5)                            | 0.000906                | 0.026657                      |
| (100, 0.5)                           | 0.000906                | 0.013402                      |
| (150, 0.5)                           | 0.000906                | 0.008979                      |
| (50, 1.2)                            | 0.002025                | 0.029876                      |
| (100, 1.2)                           | 0.002025                | 0.015225                      |
| (150, 1.2)                           | 0.002025                | 0.010161                      |
| (50, 2.3)                            | 0.300005                | 0.300494                      |
| (100, 2.3)                           | 0.300005                | 0.300123                      |
| (150, 2.3)                           | 0.300005                | 0.300055                      |

These data clearly show that the PIB is valid in the present case. Moreover, from them we can see that increasing the window size ( $r$ ) of the uneven section lowers the bound. The reason for this is that a larger window will contain more of the support of the eigenvector of  $H_{\text{free}}$  corresponding to the eigenvalue closest to our chosen  $\lambda$ , and so, the SVD can produce a better approximation to both the eigenvector and the eigenvalue. Another interesting conclusion we can draw from them is that choosing a  $\lambda$  outside of the range of the spectrum yields a smaller bound because the eigenvectors with eigenvalues at the boundary are more localized.

## 4 Conclusion and Future Work

A lower bound on  $\rho_H$  and a general algorithm for approximating the pseudospectra of infinite-volume operators with FLC were proposed in [4], and here we tested the pseudospectral inclusion bound against the free Hamiltonian with positive results.

Future work should focus on validating the PIB and the spectral gap bound against different Hamiltonians, e.g., random Hamiltonians, Hamiltonians emerging from quasicrystal systems, etc. Additionally, combining the computational tools developed for this work with the bounds on the lower norm function should enable us to prove spectral gaps of infinite-volume one-body operators.

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